Case-Disjoint Full-Field Surrogate Benchmarking for 3D Supercritical CO_2 Natural-Circulation Loops

Abstract

Full-field surrogate models for three-dimensional supercritical carbon dioxide (sCO_2) natural-circulation loops (NCLs) offer millisecond-scale field predictions in place of steady-state conjugate computational fluid dynamics (CFD) simulations that require roughly one day of compute time per operating condition (and four to five days for transient runs). However, published surrogate studies frequently rely on random node-level train–test splits or single pooled metrics that obscure generalization to unseen operating regimes. Here we establish a case-disjoint full-field surrogate benchmark from 42 steady-state 3D RANS simulations solved with the standard k–ε turbulence model and NIST `REFPROP` real-gas thermophysical properties across three heater–cooler configurations (`HHHC`, `HHVC`, and `VHVC`) and two loop scales. We evaluate 17 surrogate models (plus a conditioning ablation)—spanning gradient-boosted and randomized tree ensembles, Fourier-coordinate networks, physics-informed neural networks (PINNs), DeepONet, Latent Neural Operator, Transolver-style attention, multi-scale graph networks, 3D Fourier neural operators (`F-FNO`, `U-FNO3D`), and proper orthogonal decomposition (`POD`) reduced-order models—across three random seeds under five-fold stratified case-disjoint cross-validation and leave-one-group-out extrapolation. We show that random node splitting understates five-fold case-disjoint mean absolute error (MAE) by $6\times$ –$13\times$ on thermodynamic state fields (density, temperature, and gauge pressure) because high-capacity models memorize case-specific operating levels from neighboring mesh nodes. We further show that decomposing piezometric gauge pressure into its closed-loop isochoric thermal-pressurization offset and centered hydrostatic residual prevents small within-case pressure variance from distorting model comparisons. Across the eight exported CFD fields, no single architecture dominates every physical regime: among cross-configuration pooled models, `DeepONet` and `PINN continuity` achieve the lowest normalized error on density and temperature, `Extra Trees` best predicts case-level gauge pressure, and multiplicative feature-modulated coordinates (`Domain-informed FiLM-MLP`) outperform all competing families across both turbulence fields (k, ε) and all three velocity components (u, v, w), while within-group `POD-RBF` provides the most accurate temperature and turbulence reconstructions when interpolating on a fixed loop mesh.

Keywords: supercritical CO_2, natural-circulation loop, full-field CFD surrogate, case-disjoint validation, neural operators, proper orthogonal decomposition

1 Introduction

Natural-circulation loops (NCLs) operating with supercritical fluids remove core and decay heat without mechanical pumps by balancing buoyancy forces between heated and cooled sections against frictional and local form losses [1, 2, 3]. Supercritical carbon dioxide (sCO_2, critical point $T_c = 304.13$ K $= 30.98°$C, $p_c = 7.3773$ MPa $= 73.773$ bar) is widely used as a scaled simulant fluid for supercritical water reactor (SCWR) thermal-hydraulics and advanced compact power cycles because its lower critical pressure and temperature permit high-fidelity experiments and conjugate

computational fluid dynamics (CFD) studies near the pseudocritical locus [4, 5, 6]. Near the pseudocritical temperature $T_{\mathrm{pc}}(p)$, small variations in enthalpy or wall heat flux induce steep nonlinear variations in density ρ, specific heat c_p, viscosity μ, and thermal conductivity k, coupling core-to-wall buoyancy gradients with turbulence production and heat-transfer deterioration [7, 8, 9].

Because resolving three-dimensional conjugate heat transfer between the primary sCO_2 pipe, the solid metallic piping, and the secondary cooling-water jacket requires fine near-wall meshes and real-gas property lookups, each mesh-independent steady-state RANS simulation in our dataset requires approximately one day of compute time, while corresponding transient simulations require four to five days. Data-driven full-field surrogates—including gradient-boosted decision trees [10, 11, 12], physics-informed coordinate networks [13, 14, 15], operator-learning architectures [16, 17, 18, 19], and proper orthogonal decomposition (POD) reduced-order models [20, 21]—offer millisecond-scale inference once trained. However, three key methodological challenges limit current full-field CFD surrogate evaluations for closed natural-circulation loops:

1. **Node-random data leakage:** Splitting a pooled table of millions of spatial nodes randomly between training and test sets places adjacent mesh nodes from every operating condition into both splits, reducing the task to spatial interpolation within already-seen CFD solutions rather than generalization to unseen operating conditions.

2. **Metric distortion from closed-loop thermodynamic level offsets:** Reporting a single pooled R^2 across all test nodes conflates between-case operating-level shifts with within-case spatial profile accuracy. Conversely, in closed constant-volume loops where isochoric thermal pressurization shifts the mean gauge pressure across cases by more than 1 MPa while the within-case hydrostatic standard deviation is only ~ 1.5 kPa, naive within-case normalization produces uninterpretable pressure errors.

3. **Competing architectural inductive biases across thin-pipe loop geometries:** Closed circulation loops combine slender circular pipes ($D_i/L \sim 10^{-2}$), sharp 90° elbows, and distinct heater–cooler orientations, making it unclear when pointwise coordinate networks, operator-learning models, voxelized 3D spectral convolutions, or mesh-aligned POD bases should be preferred across thermodynamic state versus kinematic and turbulence fields.

This paper addresses all three challenges through a systematic, three-seed full-field surrogate benchmark across 42 steady-state 3D sCO_2 natural-circulation loop CFD simulations. Specifically, our main contributions are:

- **Quantification of node-split leakage versus case-disjoint generalization:** Across six classical regression and tree-ensemble baselines, we quantify the exact error inflation between 80/20 random node splits and five-fold case-disjoint cross-validation under both case-macro MAE and median case-normalized MAE (nMAE).

- **Thermodynamic analysis and two-part decomposition of closed-loop gauge pressure:** We show that between-case gauge-pressure shifts in closed small-scale loops are governed by isochoric real-gas thermal pressurization along the Span–Wagner slope $(\partial p/\partial T)_\rho \approx 0.23$ MPa K^{-1}—including a steady-state flow-regime bifurcation between stagnant conduction and buoyant circulation in `HHVC`—and introduce a pressure evaluation protocol that separates case-level mean error from centered hydrostatic residual error.

- **Comprehensive three-seed benchmark across 18 model variants and four structural groups:** We compare tree ensembles, Fourier-feature coordinate MLPs (isolating multiplicative FiLM modulation against additive input concatenation), physics-informed neural

networks, DeepONet, Latent Neural Operator, Transolver-style attention, GAMS-GNN-style graph networks, 3D factorized and U-Net Fourier neural operators (`F-FNO`, `U-FNO3D`), and within-group mesh-aligned methods (`POD-RBF`, `Nearest case`) across three random seeds (`2026`, `2027`, `2028`), supplemented by four leave-one-configuration/geometry-group-out extrapolation studies.

2 Physical Domain and 3D sCO_2 Natural-Circulation Loop Dataset

2.1 Loop Geometry, CFD Solver Configuration, and Primary Fluid Domain

The dataset comprises 42 converged steady-state 3D RANS conjugate heat-transfer simulations of closed rectangular sCO_2 natural-circulation loops executed in ANSYS Fluent and exported on cross-sectional sampling planes via ANSYS CFD-Post. All 42 simulations use the standard two-equation k–ε turbulence model with standard wall functions, gravitational acceleration acting vertically downward along the negative y-axis, $\mathbf{g} = (0, -9.81, 0)$ m s^{-2}, and NIST `REFPROP` real-gas thermophysical properties (Span–Wagner equation of state [4]) for the primary sCO_2 loop fluid. Mesh-independence verification confirmed grid convergence prior to parametric production runs; each steady-state case required approximately 24 hours of wall-clock computation (whereas transient counterparts on the same grids require 4–5 days).

The 42 cases span three heater–cooler orientation configurations and two physical loop scales (Table 1):

- **Small-scale loops (`HHHC`, 20 cases; `HHVC`, 5 cases; `VHVC`, 4 cases):** Centerline-to-centerline rectangular loop width $L_x = 0.8936$ m ($x_{\mathrm{cl}} \in [0, 0.8936]$ m) and height $L_y = 1.0000$ m ($y_{\mathrm{cl}} \in [0, 1.0000]$ m). The primary sCO_2 fluid flows inside a circular stainless-steel pipe of inner diameter $D_i = 15.0$ mm ($R_i = 7.50$ mm, $z \in [-7.50, +7.50]$ mm) and wall thickness $t_w = 2.83$ mm (outer pipe radius $R_{o,\mathrm{pipe}} = 10.33$ mm). The cooling section is a concentric tube-in-tube counterflow water jacket of length 0.660 m with annular water gap $10.33 \le r \le 12.96$ mm and outer stainless-steel jacket wall $12.96 \le r \le 15.65$ mm. In `HHHC` (horizontal heater, horizontal cooler), the heater is on the bottom horizontal leg ($y_{\mathrm{cl}} = 0$) and the cooler jacket spans $x \in [0.1168, 0.7768]$ m on the top horizontal leg ($y_{\mathrm{cl}} = 1.000$ m). In `HHVC` (horizontal heater, vertical cooler) and small-loop `VHVC` (vertical heater, vertical cooler), the cooler jacket is mounted on the right vertical leg ($x_{\mathrm{cl}} = 0.8936$ m), spanning $y \in [0.2232, 0.8832]$ m and $y \in [0.2792, 0.9392]$ m, respectively, while the heater is on the bottom leg (`HHVC`) or lower left vertical leg (`VHVC`). Each small-loop case is exported on 97 cross-sectional planes (46 constant-x and 51 constant-y planes at 0.02 m nominal spacing).

- **Large-scale loop (`VHVC`, 13 cases):** Centerline-to-centerline rectangular loop width $L_x = 3.0100$ m ($x_{\mathrm{cl}} \in [0, 3.0100]$ m) and height $L_y = 4.0800$ m ($y_{\mathrm{cl}} \in [0, 4.0800]$ m). The primary sCO_2 pipe has inner diameter $D_i = 13.84$ mm ($R_i = 6.92$ mm, $z \in [-6.92, +6.92]$ mm) and stainless-steel wall thickness $t_w = 3.58$ mm ($R_{o,\mathrm{pipe}} = 10.50$ mm). The vertical cooler jacket on the right leg ($x_{\mathrm{cl}} = 3.0100$ m, $y \in [2.220, 3.350]$ m, length 1.130 m) encloses a concentric cooling-water annulus ($10.50 \le r \le 38.93$ mm) and outer jacket wall ($38.93 \le r \le 45.80$ mm). Each large-loop case is sampled on 142 unique cross-sectional planes (61 constant-x and 81 constant-y planes at 0.05 m nominal spacing).

Each simulation is parameterized by four physical boundary/operating inputs encoded in its case identifier `{CONFIG}_P{Q}_{p}_bar_{mdot}kgs_T_{Tin}`: heater power $Q \in [10, 2500]$ W,

primary-loop operating pressure $p_{\mathrm{op}} \in \{80, 85, 90, 100\}$ bar, secondary cooling-water inlet mass flow rate $\dot{m} \in [0.010, 0.150]$ kg s^{-1}, and cooling-water inlet temperature $T_{\mathrm{in}} \in \{283.5, 300, 305, 315\}$ K (Fig. 1, where the filename typo `HHHC_P2000_80_bar_0.05kgs_T_205` is mapped to its true coolant inlet temperature $T_{\mathrm{in}} = 305$ K, consistent with its minimum fluid temperature of 305.87 K). At each sampling node (x, y, z), eight CFD target fields are exported: density ρ (kg m^{-3}), gauge pressure p_{gauge} (Pa), temperature T (°C), turbulence eddy dissipation ε (m^2 s^{-3}), turbulence kinetic energy k (m^2 s^{-2}), and Cartesian velocity components (u, v, w) (m s^{-1}). Because the raw CFD-Post cross-sectional planes slice across the full conjugate assembly, we exclude solid stainless-steel pipe/jacket wall nodes (`null` velocity/turbulence fields, $\rho \approx 7970$ kg m^{-3}) and outer cooling-water jacket annulus nodes ($\rho > 950$ kg m^{-3}, $r \geq 10.33$ mm) and deduplicate intersecting plane coordinates during preprocessing. This isolates the $5,451,341$ unique primary sCO_2 pipe-core nodes ($r \leq 7.50$ mm in the small loop and $r \leq 6.92$ mm in the large loop, $\rho \in [153.2, 892.5]$ kg m^{-3}), whose exported densities match the Span–Wagner equation of state [4, 22] at $p_{\mathrm{abs}} = p_{\mathrm{op}} + p_{\mathrm{gauge}}$ to a median relative error of 0.0008%. Figure 2 shows the case-median distributions of the eight CFD fields across the 42 simulations.

Table 1: Summary of the 42 steady-state 3D CFD cases by heater–cooler configuration and loop scale. Here $L_x \times L_y$ denotes the loop centerline dimensions and D_i denotes the inner sCO_2 pipe diameter.

Config.	Scale	Cases	$L_x \times L_y$ (m)	D_i (mm)	Q (W)	p_{op} (bar)	$\dot{m}$ (kg s^{-1})	sCO_2 nodes/case
`HHHC`	Small	20	0.894×1.000	15.00	1000–2000	80–100	0.050–0.150	97,264
`HHVC`	Small	5	0.894×1.000	15.00	10–50	90	0.050	89,771
`VHVC`	Small	4	0.894×1.000	15.00	10–40	90	0.050	89,647
`VHVC`	Large	13	3.010×4.080	13.84	250–2500	85	0.010–0.035	207,586
Total	**4 groups**	**42**	–	–	**10–2500**	**80–100**	**0.010–0.150**	**5,451,341 total**

2.2 Piezometric Gauge Pressure and Closed-Loop Isochoric Thermal Pressurization

In variable-density buoyant flows solved in ANSYS Fluent with operating density ρ_0, the exported pressure variable is the piezometric (reduced) static gauge pressure relative to p_{op}:

$$p_{\mathrm{gauge}}(\mathbf{x}) = p_{\mathrm{abs}}(\mathbf{x}) - p_{\mathrm{op}} - \rho_0 \mathbf{g} \cdot \mathbf{x}, \tag{1}$$

which satisfies the vertical hydrostatic balance $\partial p_{\mathrm{gauge}}/\partial y = (\rho_0 - \rho(\mathbf{x}))g > 0$ when $\rho(\mathbf{x}) < \rho_0$. Consequently, within any individual case c, the spatial variation of p_{gauge} around the loop is small: the within-case standard deviation $\sigma_c(p_{\mathrm{gauge}})$ is 0.15–0.39 kPa in the 1.00 m small loops and 1.45–8.79 kPa in the 4.08 m large loop, with a 42-case mean of $\bar{\sigma}_c = 1.51$ kPa.

By contrast, the case-mean gauge pressure $\bar{p}_c = \frac{1}{N_c} \sum_{i=1}^{N_c} p_{\mathrm{gauge},c,i}$ varies across cases from 1.4 kPa to 1.763 MPa (SD$(\bar{p}_c) = 337.9$ kPa, exceeding $\bar{\sigma}_c$ by 224×). Thermodynamic inspection of the raw exports resolves the physical origin of this between-case variation (Fig. 3):

- In the 13 large-loop `VHVC` cases, a gauge-pressure reference location is pinned to 0 Pa at the top of the loop ($y = 4.08$ m), so $\bar{p}_c$ remains between 1.42 kPa and 8.77 kPa.
- In the 29 small-loop cases (`HHHC`, `HHVC`, and small `VHVC`), the closed constant-volume sCO_2 domain is bounded entirely by no-slip walls without an internal pressure-outlet pin. Heating the closed loop above its initial charge state raises the loop-mean temperature $\bar{T}_c$ at constant

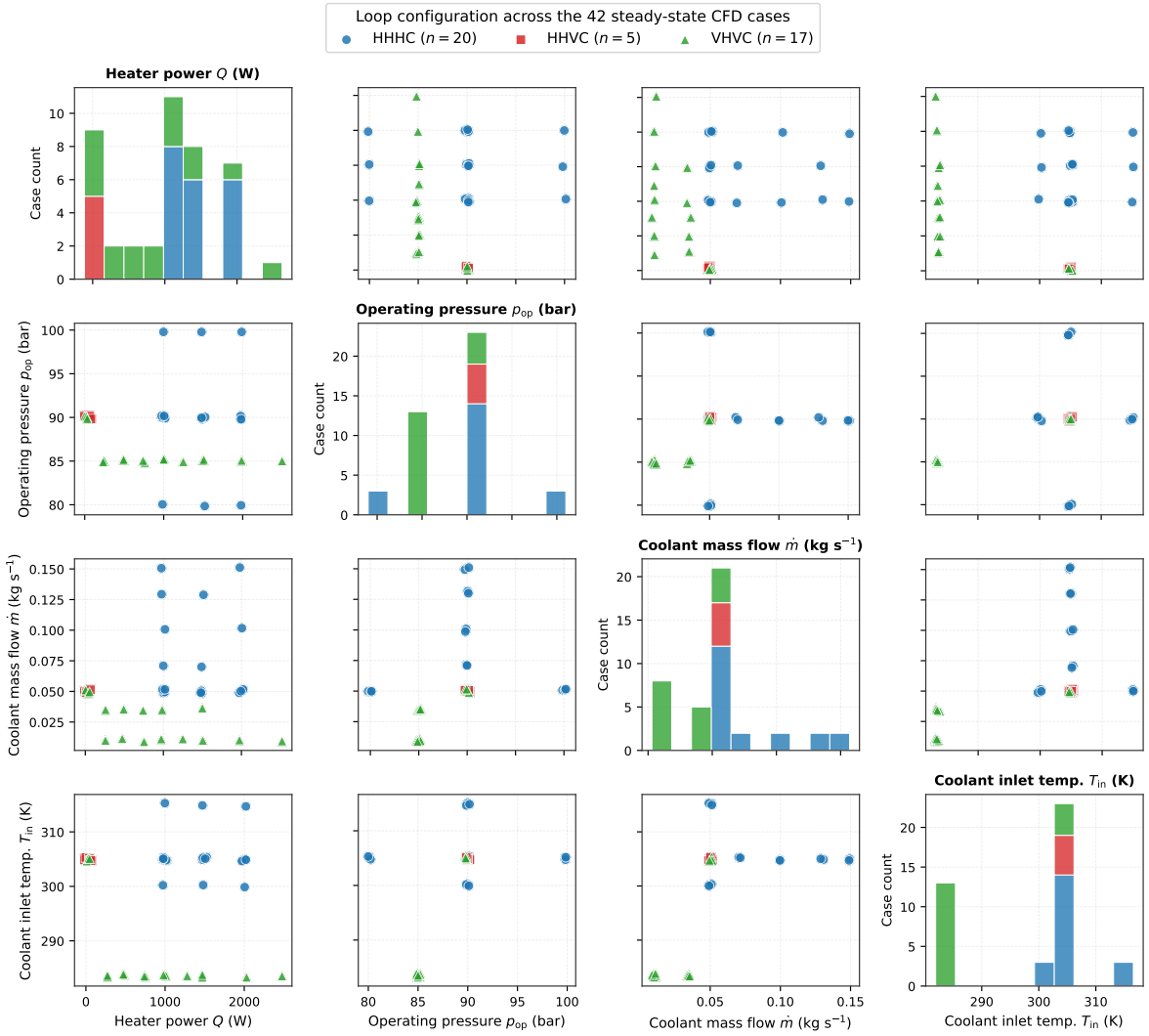


Figure 1: Operating-condition distribution across the 42 steady-state CFD cases by heater–cooler configuration (`HHHC`, `HHVC`, `VHVC`). Diagonal panels show stacked case-count histograms for heater power Q, operating pressure p_{op}, cooling-water inlet mass flow rate $\dot{m}$, and cooling-water inlet temperature T_{in} (with `T_205` corrected to 305 K).

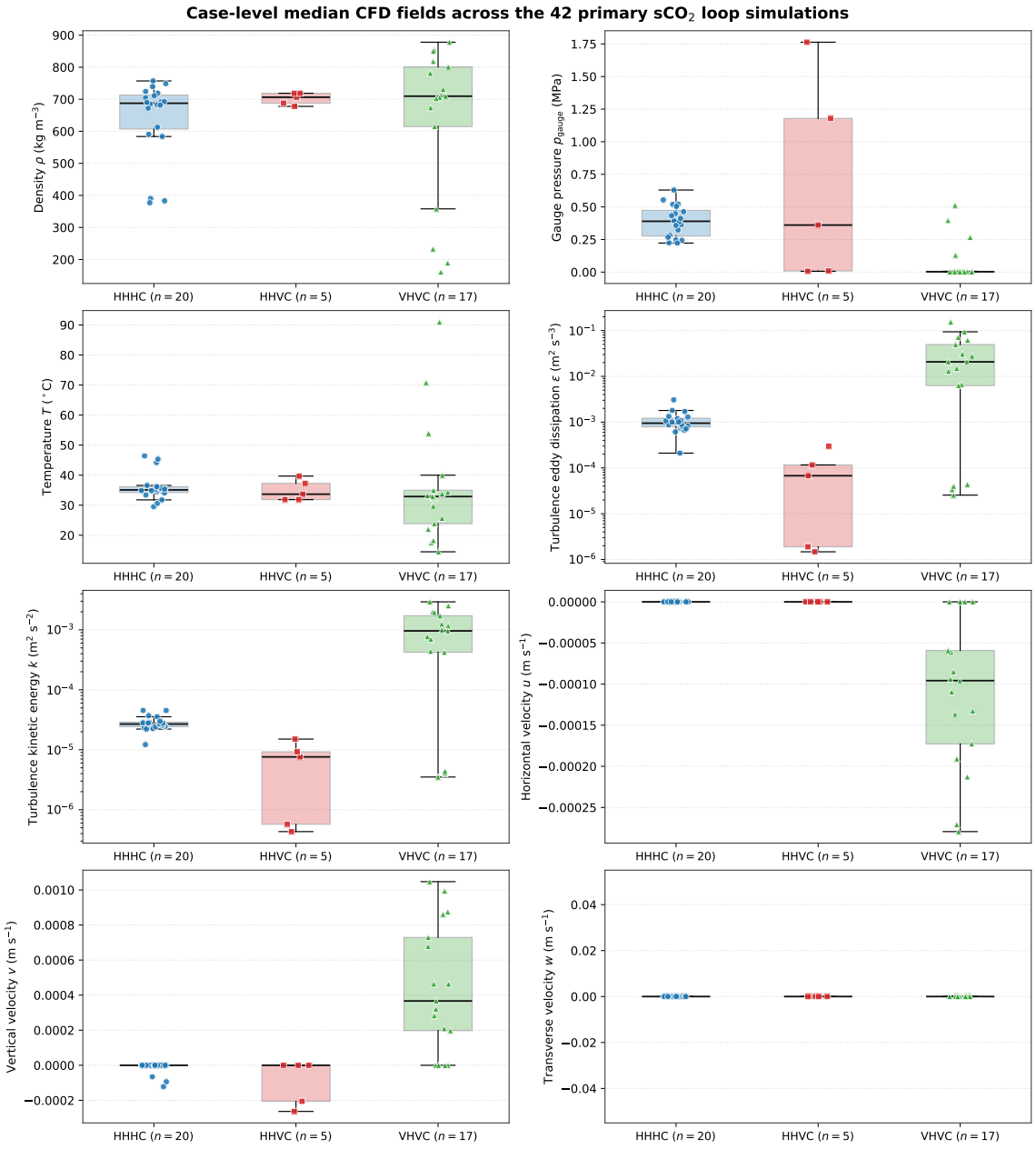


Figure 2: Case-level median distributions of the eight exported CFD fields across the 42 primary sCO_2 loop simulations (5, 451, 341 nodes), grouped by configuration (`HHHC`, `HHVC`, `VHVC`).

total fluid mass, producing an isochoric thermal-pressurization offset governed by the real-gas thermal pressure coefficient $(\partial p/\partial T)_\rho = \alpha_p/\kappa_T \approx 0.23$ MPa K^{-1} near $\rho \approx 710$ kg m^{-3}. In every case, substituting $p_{\mathrm{abs}} = p_{\mathrm{op}} + p_{\mathrm{gauge}}$ into the Span–Wagner EOS reproduces the exported nodal density ρ to within 0.00004%–0.0015%, proving that $\bar{p}_c$ is a physically active thermodynamic state offset inside the CFD solver rather than an arbitrary post-processing constant.

- In the five `HHVC` cases at $p_{\mathrm{op}} = 90$ bar and $T_{\mathrm{in}} = 305$ K (31.85°C), the apparent non-monotonicity with heater power $Q \in \{10, 20, 30, 40, 50\}$ W (Fig. 3a) reflects a physical flow-regime bifurcation in the horizontal-heater/vertical-cooler loop: the $Q = 20$ W and 30 W steady solutions converged to a near-stagnant conduction state (mean speed $\overline{|\mathbf{u}|} = 0.9$–$1.1$ mm s^{-1}, loop-mean temperature $\bar{T}_c = 31.92$–31.94°C, yielding $\bar{p}_c = 6.1$–9.2 kPa), whereas the $Q = 10$ W, 40 W, and 50 W solutions established sustained buoyant circulation ($\overline{|\mathbf{u}|} = 0.044, 0.095, 0.107$ m s^{-1}) that advected heated fluid throughout the loop ($\bar{T}_c = 33.58°\mathrm{C}, 37.09°\mathrm{C}, 39.36°\mathrm{C}$), producing monotonic isochoric pressurization of $\bar{p}_c = 0.361, 1.179$, and 1.763 MPa along the exact $(\partial p/\partial T)_\rho \approx 0.23$ MPa K^{-1} locus (Fig. 3b).

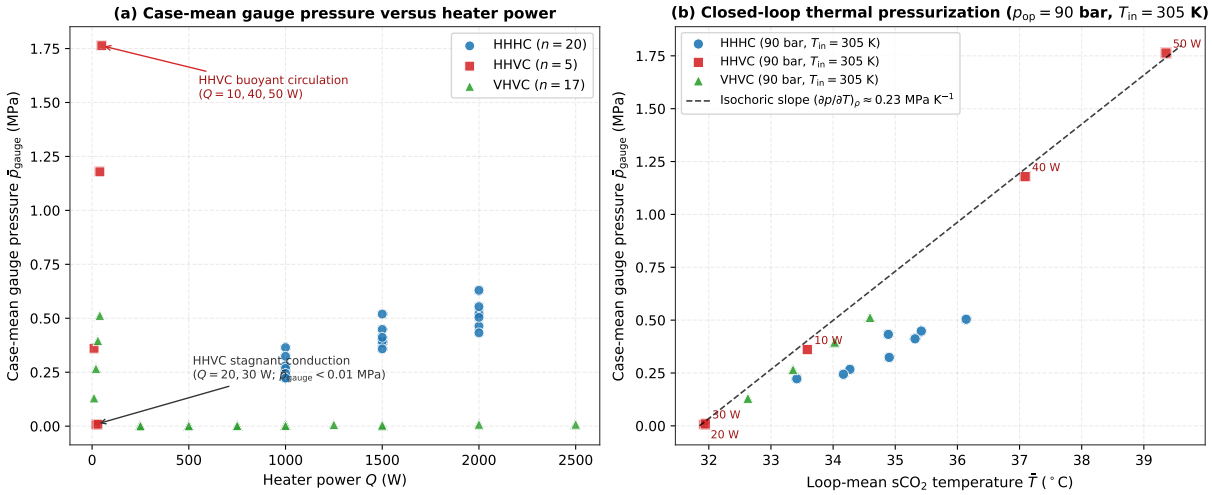


Figure 3: Physical origin of case-level gauge-pressure offsets across the 42 sCO_2 simulations. (a) Case-mean gauge pressure $\bar{p}_{\mathrm{gauge}}$ versus heater power Q, showing top-pinned large-loop `VHVC` cases near 0 MPa and closed-volume small-loop cases. (b) In the closed small-loop cases at $p_{\mathrm{op}} = 90$ bar and $T_{\mathrm{in}} = 305$ K, $\bar{p}_{\mathrm{gauge}}$ follows the Span–Wagner isochoric thermal-pressurization slope $(\partial p/\partial T)_\rho \approx 0.23$ MPa K^{-1} as a function of loop-mean temperature $\bar{T}$, explaining the difference between stagnant conduction (20 W, 30 W) and buoyant circulation (10 W, 40 W, 50 W) in `HHVC`.

3 Benchmark Methodology, Models, and Evaluation Protocol

3.1 Validation Protocols and Evaluation Metrics

Every surrogate model predicts the eight exported field variables $\mathbf{y}(\mathbf{x}, \mathbf{c}) \in \mathbb{R}^8$ at query coordinates $\mathbf{x} = (x, y, z)$ given the case-level operating and geometric descriptor vector $\mathbf{c}$ (heater power Q, operating pressure p_{op}, coolant inlet mass flow rate $\dot{m}$, coolant inlet temperature T_{in}, loop extents L_x, L_y, and one-hot configuration indicators). We evaluate three splitting protocols:

1. **Five-fold stratified case-disjoint cross-validation (primary benchmark):** The 42 cases are partitioned into five case-disjoint folds ($9, 9, 8, 8, 8$ cases) stratified by the four configuration/scale groups (`HHHC/small`, `HHVC/small`, `VHVC/small`, `VHVC/large`). All nodes of a held-out case c are unseen during training and standardization. Within each training fold, pointwise and neural models sample up to $20,000$ nodes per training case and are evaluated on 100% of the held-out case nodes ($5,451,341$ out-of-fold predictions per seed).

2. **Within-group mesh-aligned cross-validation:** Mesh-aligned reduced-order models (`POD-RBF`, `POD-GPR`, and `Nearest case`) require a shared reference node ordering and therefore train and predict strictly within the matching configuration/scale group of each five-fold split.

3. **Leave-one-configuration/geometry-group-out extrapolation:** All cases belonging to one of the four structural groups (`HHHC/small`, $n = 20$; `HHVC/small`, $n = 5$; `VHVC/small`, $n = 4$; or `VHVC/large`, $n = 13$) are held out simultaneously while pooled models train on the remaining three groups.

For each held-out case $c \in \{1, \ldots, 42\}$ with N_c nodes and ground-truth values $y_{c,i}$ and predictions $\hat{y}_{c,i}$ for a given field, we compute the case-level MAE, RMSE, within-case standard deviation σ_c, and case-normalized MAE (nMAE_c):

$$\mathrm{MAE}_c = \frac{1}{N_c}\sum_{i=1}^{N_c}|\hat{y}_{c,i} - y_{c,i}|, \qquad \mathrm{RMSE}_c = \sqrt{\frac{1}{N_c}\sum_{i=1}^{N_c}(\hat{y}_{c,i} - y_{c,i})^2}, \qquad \mathrm{nMAE}_c = \frac{\mathrm{MAE}_c}{\sigma_c}, \tag{2}$$

$$\sigma_c = \sqrt{\frac{1}{N_c}\sum_{i=1}^{N_c}(y_{c,i} - \bar{y}_c)^2}, \qquad R^2_{\mathrm{pooled}} = 1 - \frac{\sum_{c=1}^{42}\sum_{i=1}^{N_c}(\hat{y}_{c,i} - y_{c,i})^2}{\sum_{c=1}^{42}\sum_{i=1}^{N_c}(y_{c,i} - \bar{y}_{\mathrm{all}})^2}. \tag{3}$$

Across the 42 cases, we report the case-macro MAE ($\frac{1}{42}\sum_{c=1}^{42}\mathrm{MAE}_c$), case-macro RMSE ($\frac{1}{42}\sum_{c=1}^{42}\mathrm{RMSE}_c$), the median and interquartile range (IQR) of nMAE_c across cases (which prevent single-case regimes from dominating the aggregate ranking), and the node-pooled R^2_{pooled} across all $N = \sum_c N_c = 5,451,341$ nodes.

Because $\bar{\sigma}_c(p_{\mathrm{gauge}}) = 1.51$ kPa while case-mean gauge pressures span 1.763 MPa, dividing total node-level pressure error by σ_c conflates case-level thermal-pressurization offset errors with spatial gradient errors. We therefore suppress raw nMAE_c for p_{gauge} and decompose the gauge-pressure prediction into its case-mean level $\hat{\bar{p}}_c = \frac{1}{N_c}\sum_{i=1}^{N_c}\hat{p}_{c,i}$ and its centered spatial residual $\tilde{p}_{c,i} = p_{c,i} - \bar{p}_c$:

$$e_{\mathrm{level},c} = |\hat{\bar{p}}_c - \bar{p}_c|, \quad \mathrm{RMSE}_{\mathrm{res},c} = \sqrt{\frac{1}{N_c}\sum_{i=1}^{N_c}\left(\hat{\tilde{p}}_{c,i} - \tilde{p}_{c,i}\right)^2}, \quad \mathrm{nRMSE}_{\mathrm{res},c} = \frac{\mathrm{RMSE}_{\mathrm{res},c}}{\max(\sigma_c, 1\ \mathrm{kPa})}, \tag{4}$$

where the 1 kPa denominator floor prevents division by near-zero hydrostatic variance in the 1.00 m small loops ($\sigma_c = 0.15$–0.39 kPa).

3.2 Benchmark Model Suite

Table 2 summarizes the 17 benchmark models (spanning seven methodological families) plus the `Domain-informed Concat-MLP` conditioning ablation, and Table 3 lists their architecture and training hyperparameters. All neural models are trained in `PyTorch` using mixed precision (`AMP`), cosine-annealed learning rate 1.5×10^{-3}, weight decay 10^{-4}, batch size 4096, and early stopping with patience 8 over 26 epochs across three independent random seeds (`2026`, `2027`, `2028`).

To isolate the effect of multiplicative feature modulation in coordinate neural fields, `Domain-informed FiLM-MLP` and `Domain-informed Concat-MLP` share identical network depth (4 residual blocks), width (192), $B = 6$ dyadic Fourier coordinate bands $\gamma(\mathbf{x}) = [\sin(2^b\pi\mathbf{x}), \cos(2^b\pi\mathbf{x})]_{b=0}^{5}$ [14], and smooth L_1 (Huber) loss; `FiLM-MLP` modulates each hidden coordinate representation via $\mathbf{h}^{(\ell)} \odot (1 + \boldsymbol{\gamma}^{(\ell)}(\mathbf{c})) + \boldsymbol{\beta}^{(\ell)}(\mathbf{c})$ [15], whereas `Concat-MLP` concatenates $[\mathbf{x}, \gamma(\mathbf{x}), \mathbf{c}]$ at the input layer. Both `PINN data-only` and `PINN continuity` use smooth tanh activations to enable exact automatic differentiation with respect to (x, y, z), with `PINN continuity` adding a penalty of weight $\lambda_{\text{cont}} = 0.02$ on the steady compressible continuity residual $\nabla \cdot (\hat{\rho}\hat{\mathbf{u}}) = \partial_x(\hat{\rho}\hat{u}) + \partial_y(\hat{\rho}\hat{v}) + \partial_z(\hat{\rho}\hat{w})$. Voxelized 3D neural operators (`F-FNO` and `U-FNO3D`) map irregular loop coordinates onto a $24 \times 24 \times 12$ trilinear grid via inverse-distance splatting and decode back to the query mesh nodes via trilinear interpolation combined with a pointwise coordinate skip branch.

Table 2: Summary of the evaluated surrogate model families and conditioning mechanisms.

Model	Family	Conditioning & spatial representation
`Mean baseline`	Baseline	Predicts the training-fold node mean for each target
`Ridge`	Linear	ℓ_2-penalized linear regression ($\alpha = 1.0$) on $[\mathbf{x}, \mathbf{c}]$
`Random Forest`	Bagged trees	90 multi-output CART trees (max depth 18) on $[\mathbf{x}, \mathbf{c}]$
`Extra Trees`	Randomized trees	110 extremely randomized multi-output trees on $[\mathbf{x}, \mathbf{c}]$
`XGBoost`	Boosted trees	Histogram gradient-boosted trees fitted per target
`CatBoost`	Oblivious trees	Symmetric depth-8 gradient-boosted trees per target
`DeepONet`	Operator network	Branch MLP on $\mathbf{c}$ + Fourier-coordinate trunk MLP on $\mathbf{x}$
`PINN data-only`	Coordinate MLP	tanh MLP on $[\mathbf{x}, \gamma(\mathbf{x}), \mathbf{c}]$ with pure data MSE loss
`PINN continuity`	Physics-informed	Identical to `PINN data-only` + $0.02\|\nabla \cdot (\hat{\rho}\hat{\mathbf{u}})\|_2^2$ penalty
`Domain-informed FiLM-MLP`	Modulated MLP	Fourier-coordinate residual MLP with FiLM modulation + Huber loss
`Domain-informed Concat-MLP`	Ablation MLP	Identical to `FiLM-MLP` but with input concatenation $[\mathbf{x}, \gamma(\mathbf{x}), \mathbf{c}]$
`Latent Neural Operator`	Latent attention	Cross-attention from Fourier coordinates to $M = 16$ case tokens
`Transolver-style`	Physics attention	Slice-token self-attention across $S = 16$ learned physical state slices
`GAMS-GNN-style`	Graph network	Multi-scale message passing between nodes and $K = 24$ supernodes
`F-FNO`	Spectral 3D FNO	3-layer factorized 1D spectral convolutions on a $24 \times 24 \times 12$ grid
`U-FNO3D`	Multiscale 3D FNO	3D spectral layers coupled with a 3D U-Net encoder–decoder
`POD-RBF`	Aligned ROM	Within-group SVD modes (99.9% energy, ≤ 12 modes) + Gaussian RBF
`Nearest case`	Aligned lookup	Within-group nearest training case in standardized parameter space

Table 3: Principal architecture and training hyperparameters across the benchmark suite.

Model group	Hyperparameters (fixed across all five folds and three seeds)
Tree ensembles (`RF`, `ET`, `XGB`, `CB`)	`RF`: 90 trees, depth 18, min leaf 4; `ET`: 110 trees, depth 20, min leaf 3; `XGB`: 220 trees, depth 8, lr 0.07, subsample 0.8; `CB`: 320 iterations, depth 8, lr 0.08.
Coordinate & operator MLPs (`FiLM-MLP`, `Concat-MLP`, `DeepONet`, `PINNs`, `LNO`)	Width $W = 192$, depth $D = 4$, Fourier bands $B = 6$ ($\{2^0\pi, \ldots, 2^5\pi\}$), AdamW (lr $= 1.5 \times 10^{-3}$, weight decay 10^{-4}), batch size 4096, 26 epochs, patience 8.
`Transolver-style` & `GAMS-GNN-style`	Hidden width 128, $S = 16$ slice tokens / $K = 24$ graph supernodes, 4 attention heads, chunks of 2048 nodes per case, 26 epochs.
3D neural operators (`F-FNO`, `U-FNO3D`)	Grid $24 \times 24 \times 12$, spectral modes $(8, 8, 4)$, hidden channels 32, 3 spectral blocks, trilinear splatting/readout + coordinate MLP skip branch.
Mesh-aligned transfer (`POD-RBF`, `POD-GPR`, `Nearest case`)	Within-group SVD retaining 99.9% variance up to 12 modes; isotropic Gaussian kernel $K(\tilde{\mathbf{c}}, \tilde{\mathbf{c}}') = \exp(-\frac{1}{2}\|\tilde{\mathbf{c}} - \tilde{\mathbf{c}}'\|_2^2)$ with ridge $\lambda = 10^{-4}$.

4 Results and Discussion

4.1 Optimism of Node-Random Splits Versus Case-Disjoint Validation

To quantify the data leakage induced by random node-level train–test splitting, we evaluate all six classical baselines (`Mean baseline`, `Ridge`, `Random Forest`, `Extra Trees`, `XGBoost`, and `CatBoost`) under both an 80/20 random node split and five-fold case-disjoint cross-validation on the exact same $5,451,341$ primary sCO_2 nodes. Table 4 and Figure 4 report the resulting error inflation ratios (case-disjoint error divided by random-node error) under both case-macro MAE and median case nMAE.

Table 4: Optimism of random node splitting relative to five-fold case-disjoint cross-validation across all six classical baselines (`Mean baseline`, `Ridge`, `Random Forest`, `Extra Trees`, `XGBoost`, `CatBoost`). Ratios compare case-disjoint error to random-node error.

Metric	Density ρ	Pressure p_{gauge}	Temp. T	TED ε	TKE k	Vel. u	Vel. v	Vel. w
6-model median **case-macro MAE ratio**	7.09×	12.54×	6.42×	1.15×	1.74×	1.91×	1.63×	1.00×
Tree-ensemble range of **MAE ratio**	7.6–10.0×	10.5–66.0×	6.0–10.1×	1.05–1.43×	1.41–2.96×	1.80–2.63×	1.60–2.34×	0.95–1.02×
6-model median **median-nMAE ratio**	3.43×	n/a†	3.06×	1.00×	1.02×	1.39×	1.16×	1.01×

† Within-case nMAE is suppressed for gauge pressure because $\bar{\sigma}_c = 1.51$ kPa reflects only the vertical hydrostatic head.

As shown in Figure 4, whereas `Mean baseline` and `Ridge` exhibit nearly identical errors under both protocols (ratio $\approx 1.0\times$) because they lack high-frequency spatial memorization capacity, all four nonlinear tree ensembles suffer severe error inflation when evaluated on unseen cases: case-macro MAE increases by a six-model median of 12.54× on gauge pressure (up to 66.0× for `XGBoost` and 28.5× for `Extra Trees`), 7.09× on density (7.6–10.0× across tree models), and 6.42× on temperature (6.0–10.1× across tree models). Even after normalizing each case by its internal spatial standard deviation σ_c, median case nMAE is 3.43× higher on density and 3.06× higher on temperature under case-disjoint validation. Under random node splitting, high-capacity tree ensembles memorize the case-specific thermodynamic level from neighboring mesh nodes within the same simulation; under case-disjoint validation, the model must infer the entire thermodynamic state from the four scalar operating parameters $(Q, p_{\text{op}}, \dot{m}, T_{\text{in}})$.

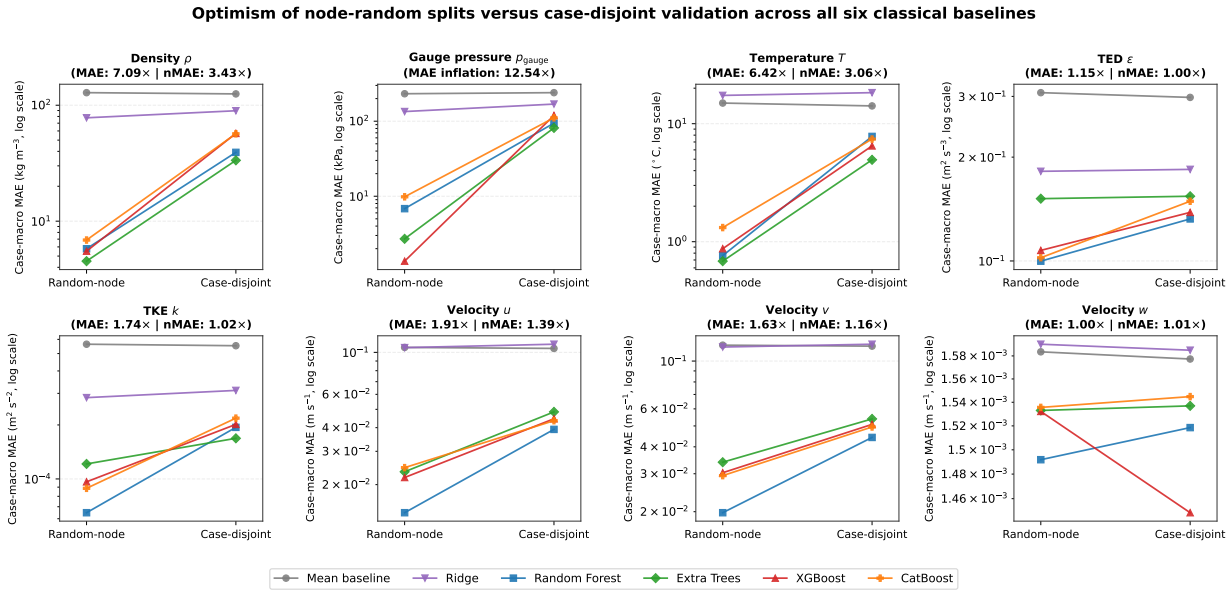


Figure 4: Comparison of case-macro MAE (log scale) under 80/20 random-node splits versus five-fold case-disjoint validation across all six classical baselines and eight CFD fields. Panel titles report the six-model median error inflation ratio under both case-macro MAE and median case nMAE.

4.2 Three-Seed Case-Disjoint Benchmark Across All Eight CFD Fields

Figure 5 presents the three-seed (2026, 2027, 2028) case-macro MAE of all 18 evaluated models normalized by the `Mean baseline` MAE across all eight CFD fields, ordered by geometric-mean relative error. Table 5 reports the detailed three-seed performance of the best cross-configuration pooled model for each field alongside the best within-group mesh-aligned transfer model where the latter achieves lower error (complete per-model metrics, IQRs, and bootstrap intervals for all 19 evaluated models appear in Supplementary Tables S1–S8).

Four key physical and architectural findings emerge from the three-seed benchmark:

1. **Thermodynamic state fields $(\rho, T, p_{\text{gauge}})$ versus kinematic and turbulence fields $(u, v, w, k, \varepsilon)$:** Table 5 shows a clear contrast between pooled R^2 and within-case normalized error (nMAE_c) for density and temperature. Pooled R^2 is high (0.862–0.936 for density and 0.736–0.806 for temperature) because the models capture the large between-case thermodynamic shifts across $p_{\text{op}} \in [80, 100]$ bar and $Q \in [10, 2500]$ W. However, because the within-case temperature and density variations inside the small-loop cases are narrow ($\sigma_c(T) \approx 0.25$–2.5°C, $\sigma_c(\rho) \approx 5$–35 kg m^{-3}), even a 2–5°C case-level mean offset produces a median case nMAE of 0.837 on ρ (`DeepONet`) and 1.350 on T (`PINN continuity`), which within-group `POD-RBF` reduces to 0.974 on T. Conversely, velocity and turbulence fields have zero or near-zero case-mean offsets and strong geometry-locked spatial gradients across the four pipe legs; here `Domain-informed FiLM-MLP` achieves low median case nMAE of 0.162 on u, 0.175 on v, 0.242 on k, and 0.192 on ε (with pooled $R^2 = 0.801, 0.861, 0.898$, and 0.569).

2. **Multiplicative FiLM modulation versus input concatenation:** Comparing `Domain-informed FiLM-MLP` against its matched ablation `Domain-informed Concat-MLP` (Fig. 5) shows that multiplicative FiLM conditioning reduces median case nMAE on six of the seven spatially varying fields: temperature (1.467 vs. 1.758), TED (0.192 vs. 0.226), TKE

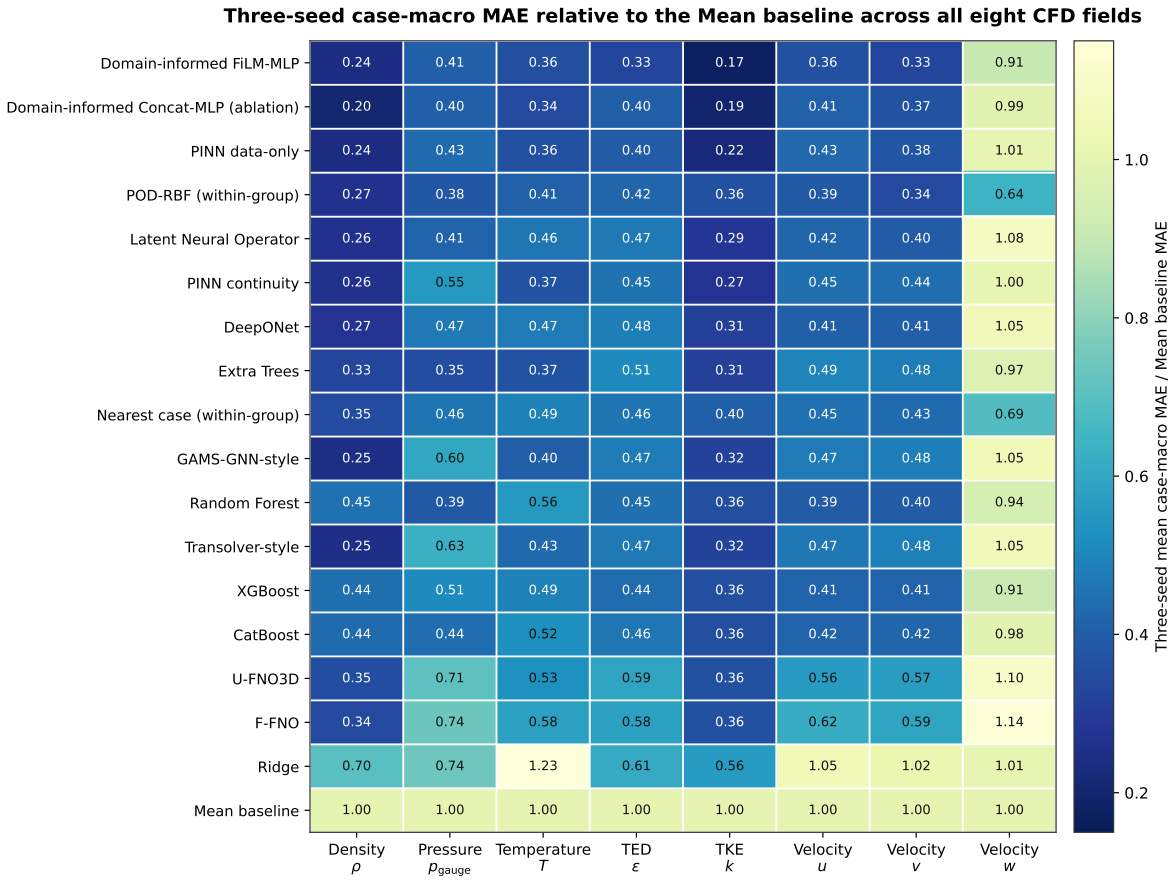


Figure 5: Three-seed mean case-macro MAE relative to the `Mean baseline` across all eight CFD fields for the 15 cross-configuration pooled models, the `Domain-informed Concat-MLP` conditioning ablation, and the two within-group mesh-aligned transfer models (`POD-RBF` and `Nearest case`), ordered by geometric-mean error ratio.

Table 5: Three-seed (2026, 2027, 2028) out-of-fold accuracy across the 42 primary sCO_2 cases. For each target, the first row gives the best cross-configuration pooled model (selected by median case nMAE for the seven spatially varying fields and by case-macro MAE for gauge pressure); where a within-group mesh-aligned model (POD-RBF or Nearest case) achieves lower error, it is listed immediately below. All MAE and RMSE values are case-macro averages.

Target (unit)	Model	Scope	Macro MAE	Macro RMSE	Median nMAE (±SD)	Pooled R^2
Density ρ (kg m^{-3})	DeepONet	Pooled 5-fold	35.1	37.1	0.837 ± 0.160	0.862
	Domain-informed Concat-MLP	Pooled (lowest MAE)	26.3	29.1	1.011 ± 0.069	0.936
Pressure p_{gauge} (kPa)	Extra Trees	Pooled 5-fold	83.2	83.7	n/a† (2.91 res. nRMSE)	0.815
	Nearest case / POD-RBF	Within-group aligned	108.1 / 90.1	108.2 / 90.2	0.0122 / 0.0483 res. nRMSE	0.776 / 0.571
Temperature T (°C)	PINN continuity	Pooled 5-fold	5.48	7.32	1.350 ± 0.044	0.736
	POD-RBF	Within-group aligned	6.11	7.42	0.974 ± 0.041	0.620
TED ε (m^2 s^{-3})	Domain-informed FiLM-MLP	Pooled 5-fold	0.0978	0.315	0.192 ± 0.020	0.569
	POD-RBF	Within-group aligned	0.127	0.369	0.177 ± 0.018	−0.575
TKE k (10^{-4} m^2 s^{-2})	Domain-informed FiLM-MLP	Pooled 5-fold	0.947	1.57	0.242 ± 0.051	0.898
	POD-RBF	Within-group aligned	2.00	2.78	0.237 ± 0.015	−0.070
Velocity u (m s^{-1})	Domain-informed FiLM-MLP	Pooled 5-fold	0.0381	0.0569	0.162 ± 0.015	0.801
Velocity v (m s^{-1})	Domain-informed FiLM-MLP	Pooled 5-fold	0.0383	0.0591	0.175 ± 0.004	0.861
Velocity w (10^{-3} m s^{-1})	Domain-informed FiLM-MLP	Pooled 5-fold	1.43	3.46	0.272 ± 0.007	0.418
	Nearest case / POD-RBF	Within-group aligned	1.08 / 1.01	2.93 / 2.55	0.185 ± 0.009 / 0.207 ± 0.008	0.551 / 0.650

† For gauge pressure, within-case nMAE is replaced by the centered spatial residual $\text{nRMSE}_{\text{res}}$ (1 kPa floor; Table S8).

(0.242 vs. 0.273), u (0.162 vs. 0.224), v (0.175 vs. 0.230), and w (0.272 vs. 0.318). Scaling hidden coordinate features multiplicatively by functions of $(Q, p_{\text{op}}, \dot{m}, T_{\text{in}})$ directly captures the Reynolds- and Grashof-number scaling of velocity and turbulence profiles around the loop, whereas additive input concatenation slightly favors global density-level regression (26.3 vs. 31.3 kg m^{-3} case-macro MAE).

3. **Gauge-pressure level versus spatial residual decomposition:** Total node-level gauge-pressure MAE (83.2 kPa for Extra Trees, 90.1 kPa for POD-RBF, 95.7 kPa for FiLM-MLP) is almost entirely governed by the case-level thermal-pressurization offset $e_{\text{level},c}$ (82.9 kPa for Extra Trees, driven by the five-case HHVC stagnant-to-buoyant bifurcation where 20 W and 30 W sit near 8 kPa while 40 W and 50 W reach 1.18–1.76 MPa). When evaluated on the centered spatial pressure gradient $\tilde{p}_{c,i} = p_{c,i} - \bar{p}_c$ (Supplementary Table S8), mesh-aligned Nearest case ($\text{nRMSE}_{\text{res}} = 0.0122$), POD-GPR (0.0457), and POD-RBF (0.0483) and tree/boosted baselines (XGBoost, 0.151) accurately recover the vertical hydrostatic profile $\partial p_{\text{gauge}}/\partial y = (\rho_0 - \rho)g$, whereas unconstrained coordinate neural networks introduce spurious high-frequency pressure oscillations ($\text{nRMSE}_{\text{res}} = 4.5$–$24.3$).

4. **Why 3D grid-splatted FNOs underperform on thin pipe loops:** F-FNO and U-FNO3D rank near the bottom of the neural models across all fields (Fig. 5). Because the pipe inner diameter (13.84–15.0 mm) occupies less than 1.5% of the 0.894×1.000 m and 3.010×4.080 m bounding boxes, splatting the loop onto a uniform $24 \times 24 \times 12$ Cartesian voxel grid collapses the entire circular pipe cross-section into a single voxel cell along the loop perimeter, erasing radial boundary-layer gradients that pointwise Fourier-coordinate networks (FiLM-MLP, DeepONet, PINNs) and within-group POD-RBF resolve directly on the unstructured mesh.

Nonparametric Friedman repeated-measures tests across the $N = 42$ cases confirm highly significant model differences on every field ($p \leq 5.7 \times 10^{-37}$; Supplementary Table S10). Paired two-sided Wilcoxon signed-rank tests with Holm step-down adjustment show that Domain-informed

FiLM-MLP achieves the lowest mean Friedman rank on horizontal velocity u (mean rank 3.69) and vertical velocity v (3.45), statistically indistinguishable from within-group POD-RBF (Holm $p = 1.00$) while significantly outperforming all tree, graph, transformer, and 3D FNO baselines ($p < 10^{-4}$).

4.3 Spatial Reconstruction Analysis and Leave-One-Group-Out Extrapolation

Figure 6 displays held-out case-disjoint CFD reference fields, surrogate predictions, and absolute error maps $|\hat{y} - y|$ for density (DeepONet on HHHC_P1500_90_bar_0.05kgs_T_300), temperature (PINN continuity on VHVC_P1000_85_bar_0.010kgs_T_283.5), and vertical velocity v (Domain-informed FiLM-MLP on VHVC_P1000_85_bar_0.035kgs_T_283.5), where the transverse radial offset from each leg centerline is magnified so the cross-sectional pipe structure is visible across all four legs. The surrogates accurately capture the axial buoyancy-driven density stratification between the heated and cooled legs and the upward riser ($v > 0$ on the left leg) versus downward downcomer ($v < 0$ on the right leg) velocity cores, while the largest pointwise errors localize within the thin thermal boundary layers inside the cooler jacket and at the four 90° elbow turns where secondary Dean vortices develop.

Finally, evaluating the 16 pooled models under leave-one-configuration/geometry-group-out extrapolation (Supplementary Table S9) demonstrates the sharp boundary between parametric interpolation and geometric extrapolation. When the small-loop VHVC/small group ($n = 4$) is held out, models trained on HHHC/small, HHVC/small, and VHVC/large have seen both the small-loop geometry (0.894×1.000 m, $p_{\text{op}} = 90$ bar, $T_{\text{in}} = 305$ K) and the vertical-heater/vertical-cooler orientation; consequently, XGBoost maintains moderate median case nMAE on density (0.980), temperature (1.609), TED (0.282), TKE (0.468), and vertical velocity v (0.686). By contrast, holding out all 13 cases of VHVC/large forces simultaneous extrapolation across a $3.4\times$–$4.1\times$ increase in loop dimensions (3.010×4.080 m), a different inner pipe diameter (13.84 mm vs. 15.0 mm), and an unseen coolant inlet temperature (283.5 K vs. 300–315 K), inflating the best density median nMAE to 3.854; similarly, holding out all 20 HHHC/small cases removes the only configuration with a top horizontal cooler leg and high-power small-loop operation (1000–2000 W), inflating the best density and temperature median nMAE to 4.601 and 9.572. Thus, cross-geometry generalization in closed natural-circulation loops requires training coverage across loop scales and heater–cooler orientations or explicit dimensionless path-coordinate normalization.

5 Conclusions

We have established a case-disjoint full-field surrogate benchmark across 42 three-dimensional steady-state sCO_2 natural-circulation loop CFD simulations ($5,451,341$ primary sCO_2 nodes). Our main conclusions are:

1. **Random node splitting severely overstates surrogate accuracy and masks closed-loop level offsets:** Across six classical baselines, 80/20 random node splits understate five-fold case-disjoint MAE by 7.09× on density, 12.54× on gauge pressure, and 6.42× on temperature (3.43× and 3.06× on median case nMAE for density and temperature). Furthermore, because closed constant-volume loops undergo isochoric thermal pressurization ($(\partial p/\partial T)_\rho \approx 0.23$ MPa K^{-1}) while local hydrostatic gradients remain near 1.5 kPa, decomposing gauge pressure into its case-level mean offset and centered spatial residual is essential for physically meaningful model evaluation.

2. **Complementary inductive biases of operator, FiLM-modulated coordinate, and POD models:** Across three independent training seeds under five-fold case-disjoint cross-

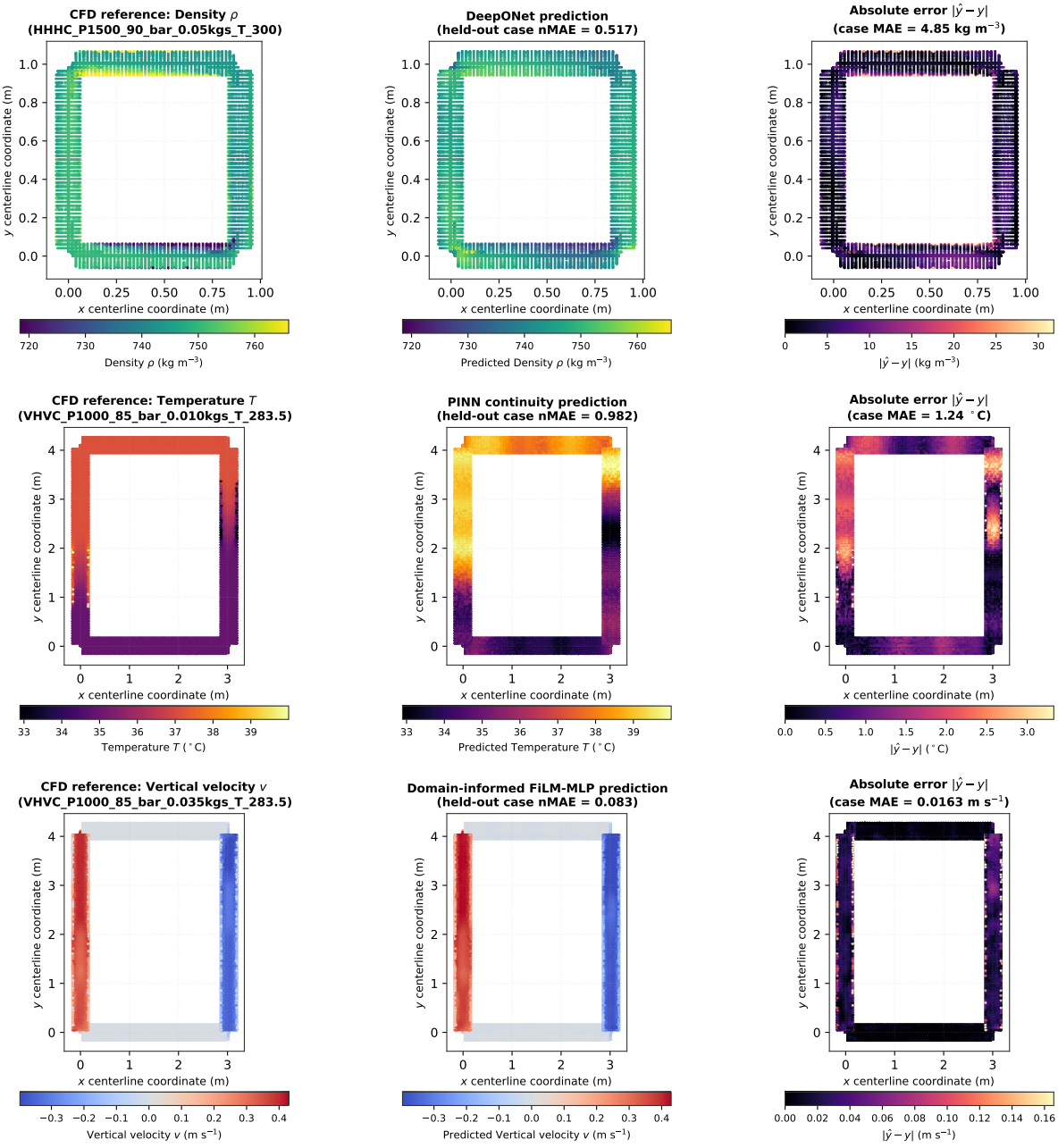


Figure 6: Held-out out-of-fold spatial field reconstructions comparing reference CFD (left), surrogate prediction (middle), and absolute error $|\hat{y} - y|$ (right) for density ρ (`DeepONet`), temperature T (`PINN continuity`), and vertical velocity v (`Domain-informed FiLM-MLP`). Transverse radial coordinates are magnified around the rectangular loop centerline (8.5× in the small loop, 24× in the large loop) to reveal cross-sectional profiles.

validation, multiplicative feature modulation (`Domain-informed FiLM-MLP`) achieves the highest accuracy on all three velocity components (nMAE $= 0.162, 0.175, 0.272$) and both turbulence fields (nMAE $= 0.192$ on ε, 0.242 on k), outperforming additive input concatenation on six of seven spatially varying fields; `DeepONet` (nMAE $= 0.837$) and `PINN continuity` (nMAE $= 1.350$) lead among cross-configuration pooled models on density and temperature; and within a fixed loop mesh, `POD-RBF` achieves the lowest within-case nMAE on temperature (0.974), TED (0.177), and TKE (0.237). By contrast, uniform 3D grid-splatted spectral operators (`F-FNO`, `U-FNO3D`) lose radial boundary-layer resolution due to the slender pipe aspect ratio ($D_i/L < 1.5\%$).

3. **Interpolation versus cross-scale geometric extrapolation:** Leave-one-group-out experiments confirm that pooled coordinate and tree models transfer effectively across heater orientations when the loop scale and operating pressure have been seen during training (holding out `VHVC/small`), but degrade sharply when simultaneously extrapolating across a $3.4\times$–$4.1\times$ change in loop dimensions, pipe diameter, and coolant inlet temperature (holding out `VHVC/large` or `HHHC/small`), motivating dimensionless arc-length and radial coordinate parameterizations for multi-scale loop surrogates.

Data and Code Availability

All preprocessing, CPU and GPU benchmark training, statistical testing, and figure/table generation scripts, along with the complete three-seed metric tables and supplementary material (Tables S1–S10), are publicly available at https://github.com/Barghav777/SCWR_3D.

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